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Superhydrophobic, superamphiphobic and SLIPS materials as anti-corrosion and anti-biofouling barriers

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Nature-inspired interfacial materials with special wettability have received considerable attention owing to their superior properties and promising multifunctional applications. Owing to the unique interfacial phase contacts and liquid-repellent property, the interfacial materials with special wettability provide a great potential for corrosion inhibition and biofouling suppression. Thus, this brief review and perspective study provides an overview about the characteristics, fabrication and recent advances of three types of interfacial materials with special wettability, namely superhydrophobic, superamphiphobic and slippery liquid-infused porous surface (SLIPS) toward anti-corrosion and anti-biofouling applications. The bottlenecks and future research priorities of the functional interfacial materials with special wettability were pointed out to accelerate the comprehensive understanding and the development of this research field.

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1. Introduction

Corrosion and biofouling can be defined as chemical-physical-electrochemical interactions between solid materials and surrounding environments, which are widespread and inevitable

problems happening in all fields of the world's infrastructures and equipment including highways, bridges, aerospace, port wharfs, offshore platforms, buildings, industrial manufactures, and military facilities (Fig. 1). Globally, annual corrosion costs account for 3–5% of the gross domestic product (GDP) of industrialized countries.^{1–3} The corrosion and biofouling of different kinds of materials result in performance degradation, huge economic loss, ecologically environmental damage or even catastrophic safety accidents.

A variety of strategies have been reported and utilized to improve the anti-corrosion and anti-biofouling performance, such as element alloying,^{4,5} sacrificial anodes,^{6,7} corrosion

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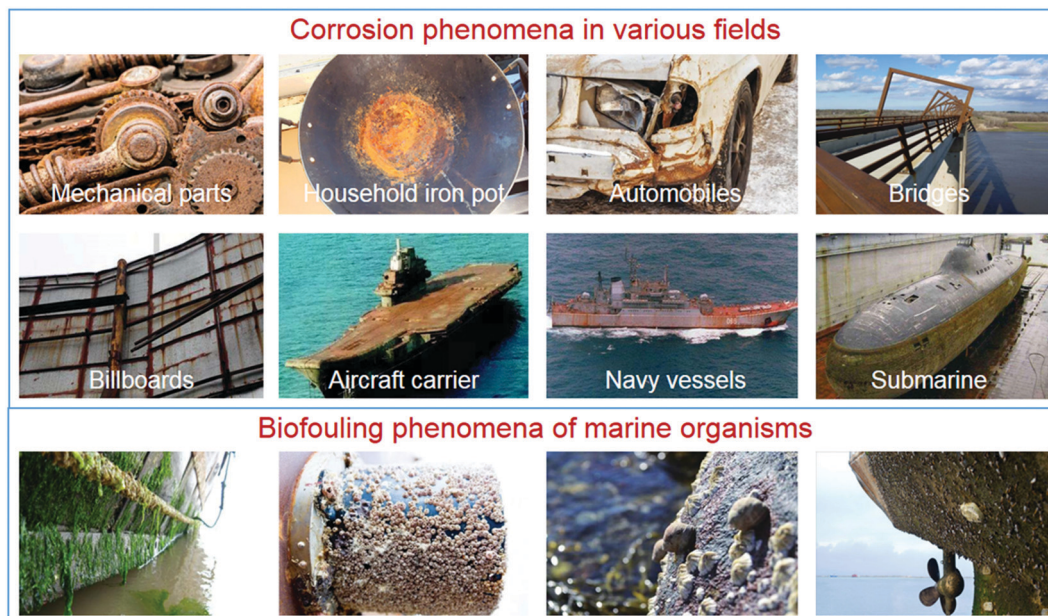


Fig. 1 Corrosion and biofouling phenomena in various fields.

inhibitors,^{8,9} surface treatments,^{10,11} organic coatings,^{12,13} nano-composite coatings,^{14,15} biocidal agents,^{16,17} photo-catalytic sterilization,^{18,19} and anti-fouling coatings.^{20,21} However, numerous challenges remain for improving the applicability, long-term protective performance, and for the reduction of their hazardous effects on environment and human health. Thus, it is highly desirable to design and fabricate potential protective materials with high effectiveness, low cost, environmental compatibility, and long-term durability and stability towards anti-corrosion and anti-biofouling applications.

Surface wettability is one of the most significant properties of solid surfaces playing an important role in determining the chemical and physical performance. Inspired by the unique surface/interface properties of natural organisms, functional interfacial materials with special wettability (superhydrophobic, superamphiphobic, ultra-slippery *etc.*) provide promising applications in oil-water separation,^{22,23} self-cleaning,^{24,25} anti-icing,^{26,27} drag reduction,^{28,29} energy harvesting,^{30,31} water collection,^{32,33} surface enhanced Raman spectroscopy,^{34,35} *etc.* It is particularly worth mentioning that typical interfacial air-liquid-solid contacts of these materials can be considered as alternative means for improving the corrosion resistance^{36–38} and the biofouling suppression performance^{39,40} under different environmental circumstances.

In this review and perspective, we mainly focus on three types of functional interfacial materials with special wettability, *viz.* superhydrophobic, superamphiphobic and slippery liquid-infused porous surface (SLIPS) materials. The brief introduction, fabrication strategies and functional protective applications as anti-corrosion/anti-biofouling barriers are summarized and discussed. Finally, we summarized and pointed out the bottlenecks and future research prospects of the interfacial materials with special wettability.

2. Superhydrophobic materials

2.1. Superhydrophobic phenomena in nature

The contact angle value is determined by the tangent of the liquid droplet at the constant point between the solid-liquid-air phases with the solid phase surface at the three-phase boundaries. Owing to the difference of wetting/non-wetting behaviors and the water contact angle (WCA) values, the solid surfaces can be classified into four categories. WCA in the range of $0^\circ < \theta < 10^\circ$, $10^\circ < \theta < 90^\circ$, $90^\circ < \theta < 150^\circ$, and $\theta > 150^\circ$ can be termed as superhydrophilic regime, hydrophilic

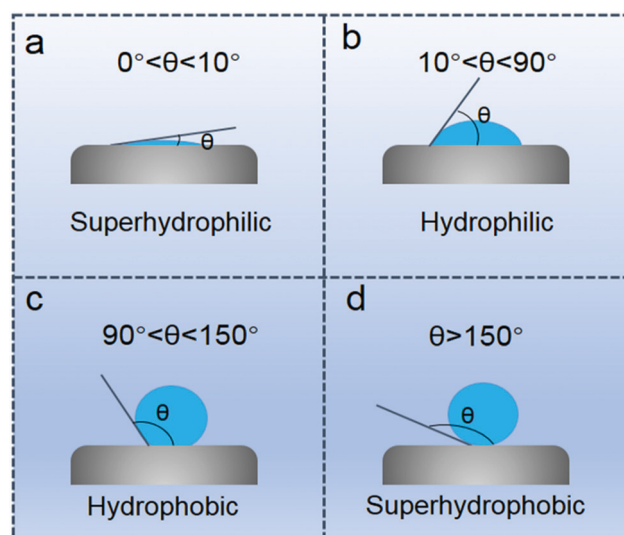


Fig. 2 Schematic of (a) superhydrophilic, (b) hydrophilic, (c) hydrophobic, and (d) superhydrophobic surfaces with different wetting/non-wetting states.

regime, hydrophobic regime, and superhydrophobic regime, respectively (Fig. 2).

A superhydrophobic surface is a surface with a WCA higher than 150° and a sliding angle (SA) lower than 10° , describing a state of perfect non-wetting contact. Superhydrophobic surfaces are inspired from the typical non-wetting behavior of the interface of natural organisms (Fig. 3), such as lotus leaves,⁴¹ water strider legs,⁴² peanut leaves,⁴³ rose petals,⁴⁴ rice leaves,⁴⁵ taro leaves,⁴⁶ butterfly wings,⁴⁷ mosquito eyes,⁴⁸ cicada wings,⁴⁹ and gecko feet.⁵⁰ German botanists Barthlott and Neinhuis⁵¹ discovered the self-cleaning properties of lotus leaf surfaces, *viz.* “Lotus effect”. It was believed that this unique property is based on the surface roughness caused by the micrometer-scale papillae and the epicuticular wax.⁵² However, the fundamental mechanism remains unclear. Jiang *et al.*⁵³ first reported the hierarchical micro-nano structures in a lotus leaf systematically. The binary synergistic effects of micro-papillae and branch-like nanostructures both contribute to the surface superhydrophobicity.

The artificial design and fabrication of superhydrophobic materials for various applications attract significant attention of both scientists and engineers. An accelerating increase in the number of studies published each year from 2000 to 2020 was found by searching in the ISI Web of Science core collection for the topic of “superhydrophobic” (Fig. 4). The research records of superhydrophobic materials verified the significance and frontiers of this research fields.

2.2. Fabrication and corrosion/biofouling applications

As previously introduced, the surface roughness and chemical composition play key roles for achieving superhydrophobic materials. Over the past two decades, researchers have artificially fabricated numerous superhydrophobic materials based

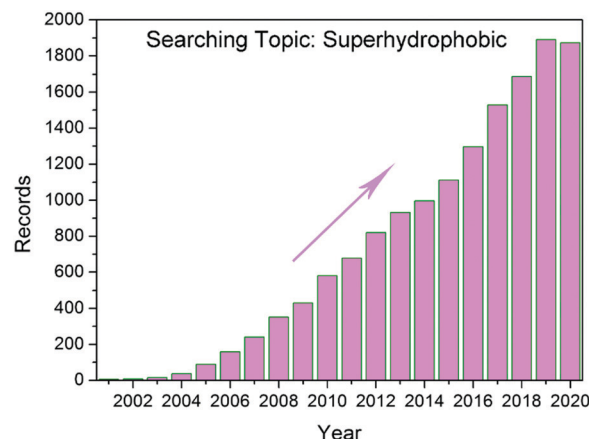


Fig. 4 Number of publications indexed in Web of Science from 2000 to 2020 when searched using “superhydrophobic” as the keyword.

on the following two principles: (1) modifying a rough surface with low surface energy substances and (2) creating roughness in hydrophobic materials. A variety of materials and strategies (Fig. 5) have been utilized to construct superhydrophobic materials towards anti-corrosion and anti-biofouling applications. A brief description of the commonly employed fabrication techniques and the corresponding corrosion/biofouling resistance behaviors are presented in the following.

Sun *et al.*⁵⁴ fabricated a superhydrophobic surface with controllable periodic structures on an AISI304 stainless steel substrate through picosecond laser texturing followed by silicone sol coating. The anti-biofouling performance was studied by seawater immersion for five weeks. The results demonstrated that the laser-textured superhydrophobic surfaces exhibit significant anti-biofouling effects (nearly 50% decrease in

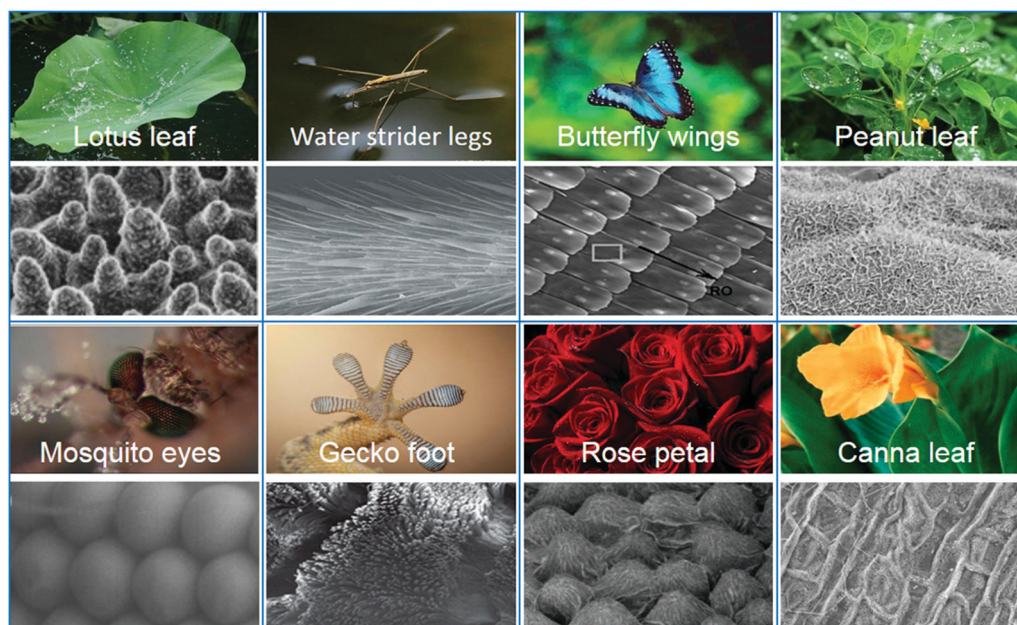


Fig. 3 Superhydrophobic phenomena of interfaces of natural organisms.

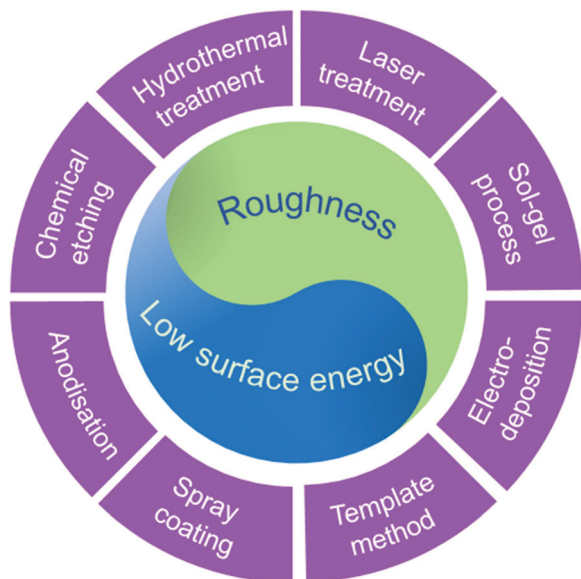


Fig. 5 Fabrication strategies of artificial superhydrophobic materials.

the average microbe attachment area ratio) compared to the bare stainless steel plate. In addition, the authors indicated that more abundant micro-nano structures induced by laser led to the better anti-biofouling performance. Cui *et al.*⁵⁵ fabricated superhydrophobic polyvinylidene fluoride (PVDF)/stearic acid (SA) nano-fiber coatings on aluminum substrates for anti-corrosive protection *via* a controllable electrospinning technology. The as-fabricated superhydrophobic PVDF/SA nano-fiber coating exhibited larger impedance and lower corrosion current density, even for a 30 days immersion in a 3.5 wt% NaCl corrosive solution, demonstrating the superior anti-corrosion property and application potentials. Syed *et al.*⁵⁶ proposed a layer-by-layer (LBL) spin-assembled polyaniline-silica composite/tetramethylsilane-functionalized silica nanoparticles (PSC/TMS-SiO₂) coated on 316 stainless steel with the combined effects of superhydrophobicity and enhanced anti-corrosion ability. The electrochemical results show that the polarization resistance (R_p) of the superhydrophobic PSC/TMS-SiO₂ coating is 126.48 kΩ cm², which is nearly six times higher than the uncoated sample (22.97 kΩ cm²), indicating a significant enhancement of the corrosion resistance of the superhydrophobic coating. In our previous research studies, we have developed numerous superhydrophobic materials *via* chemical etching,⁵⁷ anodisation,⁵⁸ hydrothermal,^{59,60} one-step electrodeposition,^{61,62} spraying coating,⁶³ physical deposition,⁶⁴ *etc.* The corrosion resistance and biofouling inhibition (sulfate-reducing bacteria, *Chlorella vulgaris*, *etc.*) properties (Fig. 6) of the as-prepared superhydrophobic materials have been evaluated and analyzed, demonstrating significantly improved anti-corrosion and anti-biofouling performance.

The anti-corrosion mechanism of superhydrophobic materials can be summarized as follows. The air-liquid-solid three-phase interfacial contact of superhydrophobic materials exhibits a typical Cassie-Baxter state with a low liquid-solid contact area and high liquid-air contact area. The remarkably enhanced corrosion resistance of superhydrophobic materials

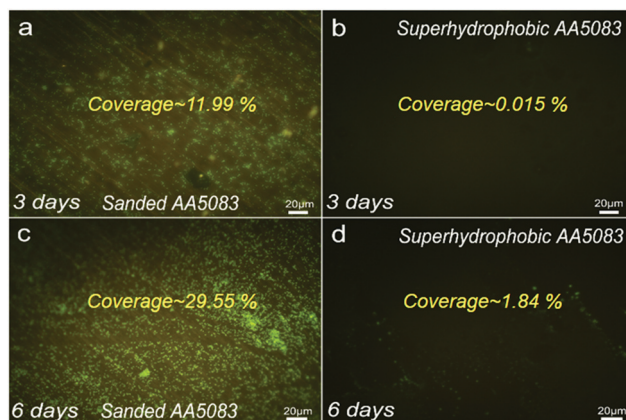


Fig. 6 Fluorescence micrographs of (a and c) bare and (b and d) superhydrophobic 5083 aluminum alloy specimens after 3 days and 6 days sulfate-reducing bacterium (SRB) culture medium immersion experiments. (Reproduced from ref. 57 with permission from Elsevier, Copyright 2019, License number 5106341263993.)

is attributed to the trapped air layer within the surface micro-nano hierarchical structures. The air layer provides inherent insulation, and hinders the direct contact between the corrosive media and underlying substrates. As a result, the diffusion and penetration of electron and corrosive ions between the substrate and electrolyte can be greatly inhibited, or even prevented. In addition, the Laplace pressure created in the superhydrophobic surface is beneficial for the improvement of corrosion resistance by preventing the entrance of the electrolyte. As for anti-biofouling, the typical Cassie-Baxter contacts of the superhydrophobic materials significantly reduce the attachment sites of microorganisms. In addition, the sharp nano-structured surface can also damage the bacterial membrane, and break the cell integrity of the bacteria, *viz.* killing bacteria by physically rupturing/stretching the bacteria cell wall.^{65,66} Thus, superhydrophobic materials exhibit greatly enhanced anti-corrosion and anti-biofouling properties.

However, the anti-corrosion and anti-biofouling activities of most of the current superhydrophobic materials are short-lived because the superhydrophobic state is a metastable interfacial state. The transformation from the Cassie-Baxter contact to the thermodynamically preferred Wenzel contact is easy to trigger during the utilization process under different circumstances. In addition, the superhydrophobic materials suffer from poor durability when subjected to the degradation of low surface energy molecules, external forces, mechanical vibration, evaporation, *etc.*, losing the non-wetting water repellency properties. The prolonged immersion of superhydrophobic materials in marine environment leads to the loss of air cushion and subsequently accelerated the localized corrosion and microorganism adhesion.

3. Superamphiphobic materials

It is well known that the liquid repellency performance of the anti-wetting surface is highly dependent on liquid properties.

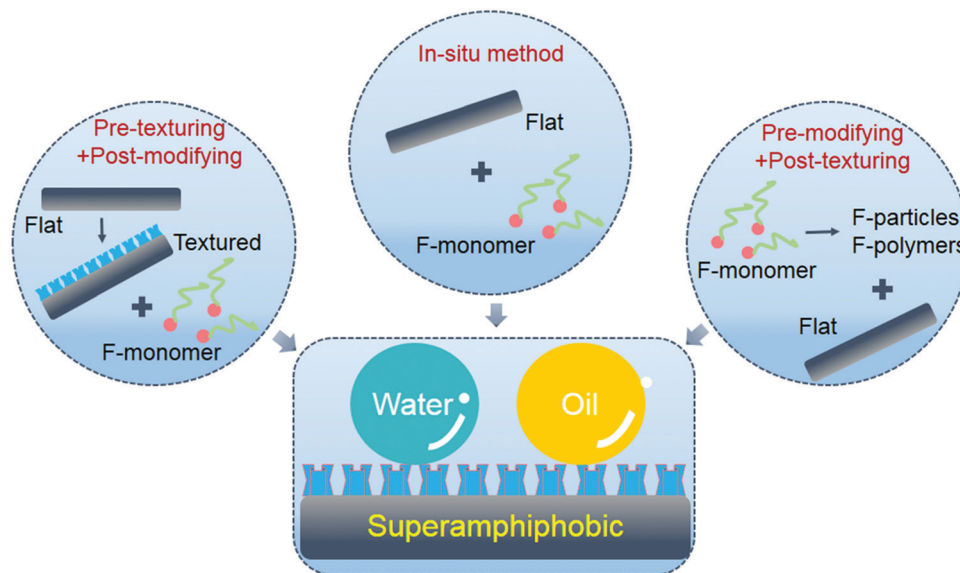


Fig. 7 Fabrication strategies of superamphiphobic materials.

The air cushion and superhydrophobic properties collapsed when contacting with low surface tension liquids, such as glycerol, diiodomethane, ethylene glycol, toluene, *n*-hexadecane, *n*-tetradecane, and *n*-decane.⁶⁷ Differently from superhydrophobicity, a surface is deemed as superamphiphobic if it exhibits super-repellency towards probing liquids with a contact angle (CA) higher than 150° for not only water but also low surface tension liquids, *e.g.* oils, organic solvents and surfactant solutions.⁶⁸ It is still challenging to construct superamphiphobic surfaces because of the extremely low surface tension of organic liquids compared to that of water. The fabrication of the superamphiphobic surface requires carefully designed surface architectures (*e.g.*, overhanging, re-entrant and doubly re-entrant)^{69–71} and materials with extremely low surface energy.

In general, the design and fabrication of superamphiphobic materials can be classified into three strategies including “Pre-texturing + Post-modifying”, “Pre-modifying + Post-texturing”, and “*In situ* achievement”,⁷² as shown in Fig. 7. Recent years, some techniques, such as template method,⁷³ oxidation process,^{74,75} sol-gel,^{76,77} electrostatic dusting,⁷⁸ and spray-coating,^{79–81} were reported to construct superamphiphobic surfaces on different metallic substrates for anti-corrosion and anti-biofouling applications. In our previous research study, we used chemical texturing + fluorosilane modification⁸² and one-pot hydrolytic co-condensation⁸³ methods to prepare superamphiphobic materials on aluminum alloys. The as-fabricated superamphiphobic surfaces have a wide range of liquids repellency to water, glycerol, ethylene glycol, peanut oil and even *n*-hexadecane (surface tension: 27.5 mN m^{−1}), outstanding self-cleaning, chemical stability and long-term anticorrosion performance.

Recent years, Zhang *et al.* have created a series of superamphiphobic coatings towards corrosion inhibition based on SiO₂ nanoparticles,^{84,85} natural attapulgite,⁸⁶ palygorskite (PAL)

nanorods and organosilanes,⁸⁷ multiwalled carbon nanotubes (MWCNTs),⁸⁸ *etc.*, The remarkably strengthened anti-corrosion performance of superamphiphobic materials (Fig. 8) compared to most of the superhydrophobic materials can be attributed to the exceptional superhydrophobic and superoleophobic properties with stable trapped air cushion among hierarchical micro/nano structures, which efficiently impedes the penetration of corrosive ions and prevents the infiltration of NaCl microdroplets to underlying metallic substrates. Thus, the superamphiphobic materials show immense potential applications in long lasting corrosion protection and biofouling suppression. However, it is still challenging to achieve superamphiphobic materials with accurate structural design, oil-repellency, ultra-low sliding angles, and mechanical robustness *etc.* In addition, the anti-biofouling performance of superamphiphobic materials is rarely been evaluated and reported, which needs increasing attention in the near future.

4. Slippery liquid-infused porous surface (SLIPS)

The formation of superhydrophobic and superamphiphobic materials is based on the Cassie–Baxter interfacial contact state. Air cushion is trapped within the macro/micro/nano rough structures, impeding the contact of the solid surface and liquid droplets, enabling various droplets rolling off easily. However, the trapped air cushion cannot stand up to pressure and mechanical damage. The scratching and pressure increasing results in the irreparable degradation of superhydrophobic and superamphiphobic properties. Recent years, slippery liquid-infused porous surface (SLIPS) attracts considerable attention of scientists due to the possibility of addressing the above-mentioned limitations. SLIPS is inspired from the functional property of the *Nepenthes* pitcher plant, as shown in

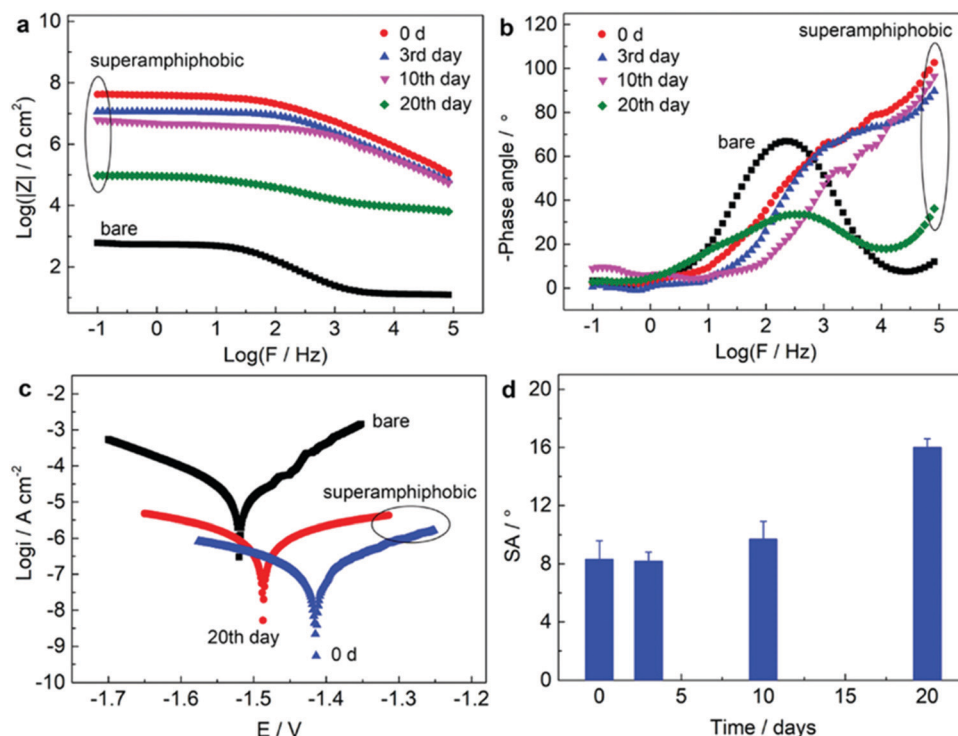


Fig. 8 Long term corrosion resistance of superamphiphobic coating. (Reproduced from ref. 84 with permission of Wiley, Copyright 2019, License number 5106350528738.)

Fig. 9. The microstructures of the *Nepenthes* pitcher plant can lock-in an intermediary liquid, generating a continuous liquid cushion repelling the oil on the feet of the insects. As a result, the insects can hardly stand on the rim of a pitcher plant and will slide down from the rim to the digestive juices filled bottom easily.^{89,90} Aizenberg *et al.*⁹¹ first reported a SLIPS by infiltrating a functionalized porous/textured solid with a low-surface-energy, and chemically inert liquid to form a physically smooth and chemically homogeneous lubricating film on the surface of the substrate. The SLIPS exhibits unique characteristics of low sliding angles, extreme pressure stability, optical transparency, negligible contact angle hysteresis, instantaneous and repeatable self-repairing, *etc.*, providing a

straightforward and versatile solution for liquid repellency and corrosion/biofouling resistance.

The fabrication strategies of SLIPS mainly include surface texturing, low surface energy modification and liquid infusion. The design principles follow several criteria such as (1) macro/micro/nano porous structures required for lubricant locking, (2) the lubricant liquid must be immiscible with repelling liquids, and (3) the lubricant liquid must wet the solid substrate more easily than repel liquids. During the past five years, some techniques such as anodisation,⁹² wire electrical discharge machining,⁹³ electrochemical texturing,⁹⁴ and thermally induced phase separation,⁹⁵ followed with liquid infusion have been reported to construct SLIPS on metallic substrates for corrosion resistance and biofouling inhibition. For instance, Long *et al.*⁹⁶ selected natural attapulgite (APT) with an existing micro-nano porous structure as the raw material to fabricate a robust superhydrophobic surface *via* the spraying coating technique. Then, the SLIPS was achieved by infusing silicone oil within the superhydrophobic surface. The fabricated SLIPS shows superior liquid repellency, anti-corrosion, anti-icing performance with intensified durability compared with the superhydrophobic surface. Qiu *et al.*⁹⁷ developed a nanoscale $\text{Cu}(\text{OH})_2$ bundle cluster structure *via* a facile electrodeposition-oxidation strategy on CuZn metal surface. After the surface modification with dodecanethiol and perfluorinated lubricant (PFL) oil infusion, the SLIPS was finally achieved. The results confirm that SLIPS has excellent corrosion inhibition ability toward the underneath metal, even when immersed in seawater for a long time. Moreover, SLIPS can effectively suppress the

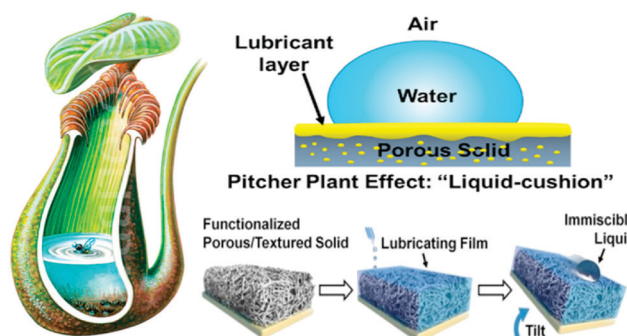


Fig. 9 Schematics of the Pitcher plant inspired slippery liquid-infused porous surface (SLIPS). (Reproduced from ref. 90 with permission of Elsevier, Copyright 2020, License number 5106360091528.)

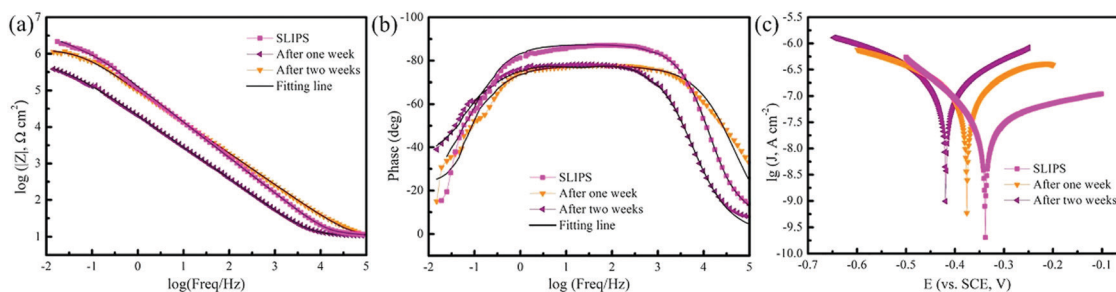


Fig. 10 Long term corrosion resistance of SLIPS materials. (Reproduced from ref. 98 with permission of Elsevier, Copyright 2018, License number 5106360521820.)

attachment of SRB and inhibit the bio-corrosion process. In addition, Xiang *et al.*⁹⁸ and Wang *et al.*⁹⁹ demonstrated that the SLIPS materials presented superior long term corrosion resistance and biofouling inhibition performance, respectively, as shown in Fig. 10 and 11. Thus, the SLIPS materials supply new promising alternatives as long-lasting protective barriers towards anti-corrosion and anti-biofouling. Further theoretical understanding of the protective mechanisms of SLIPS materials need to be established.

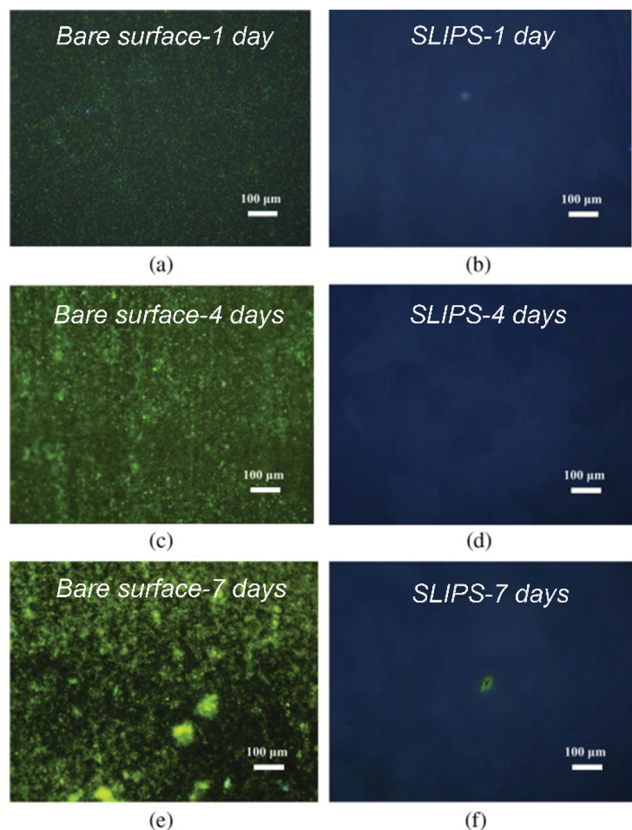


Fig. 11 Fluorescence micrographs of (a, c and e) bare aluminum surface and (b, d and f) SLIPS-coated aluminum after exposure in sulfate reducing bacterium (SRB) medium for 1 day, 4 days, and 7 days. (Reproduced from ref. 99 with permission of Elsevier, Copyright 2015, License number 5106361131254.)

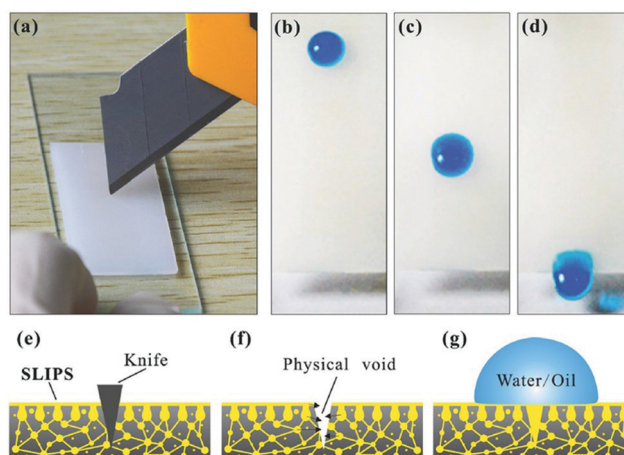


Fig. 12 Self-repairing ability of SLIPS materials. (Reproduced from ref. 100 with permission of Wiley, Copyright 2017, License number 5106350916450.)

More importantly, Yong *et al.*¹⁰⁰ and Meng *et al.*¹⁰¹ reported that the construction principle of SLIPS endowed the surface with the self-healing/self-repairing ability. Once the SLIPS materials are damaged by scratching, cutting or abrasion, the liquid-repellent property of the SLIPS can be spontaneously restored rapidly without any external forces, as shown in Fig. 12. The surface energy-induced capillary action triggers the freely flowing of the infused liquids trapped within the micro/nano porous structures, re-infusing the physical damaged area of the SLIPS. Some recently published research studies^{102–104} further demonstrated that the outstanding self-healing/self-repairing properties of SLIPS materials were due to the unique lubricant self-replenishing performance. We believe that the self-healing and lubricant-controlled release SLIPS materials will have significant applications for long-term corrosion and biofouling resistance in different environments.

5. Conclusion and perspectives

Over the past two decades, the research field of functional interfacial materials with special wettability towards anti-corrosion and anti-biofouling applications has seen a significant progress. This review summarizes the critical aspects related to the utilization of superhydrophobic, superamphiphobic and

SLIPS materials for corrosion resistance and biofouling inhibition. However, moving superhydrophobic, superamphiphobic and SLIPS materials from the laboratory scale to real world applications is a difficult issue. Up to date, most of the developed interfacial materials with special wettability are not yet readily utilized in industrial fields. Thus, lots of bottlenecks and challenges need to be addressed. The perspectives about the possible future research directions are presented but not limited to the following.

Developing facile, efficient and applicable fabrication methods and techniques for the large-scale fabrication, particularly fluorine-free, environment-friendly and low-cost chemicals as raw materials for the preparation process.

Creating new hierarchical structures or functional reagents (inhibitor, bactericide, biocides, deicing agents, 2D materials, *etc.*) modified interfacial materials with broadened functional applications (high transparency, solar-driven interfacial evaporation, *etc.*)

Design and fabrication of superhydrophobic and superamphiphobic surfaces and coatings with mechanical/chemical/thermal robustness and long-term durability and weather resistance.

Stimulus-responsive reversible/controllable wetting transition of interfacial materials and smart self-healing (microcapsules, shape memory polymer, *etc.*), self-reporting and self-adapting superhydrophobic and superamphiphobic materials.

In situ observing/monitoring of anti-corrosion and anti-biofouling and multifunctional integration applications of interfacial materials with special wettability from laboratory to real world applications.

Conflicts of interest

The authors declare no conflicts of interest.

Acknowledgements

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